

Math 290

Practice Quiz 2

1. Short response.

(a) Evaluate the following proposed theorem and proof.

Theorem: Let $x, y \in \mathbb{Z}$. If $x + y$ is odd, then xy is even.

Proof. Let $x, y \in \mathbb{Z}$. Assume that $x + y$ is odd. Then $x + y = 2k + 1$ for some $k \in \mathbb{Z}$.

Case 1: Assume that x is odd. Then $x = 2\ell + 1$ for some $\ell \in \mathbb{Z}$. Now

$$y = (x + y) - x = (2k + 1) - (2\ell + 1) = 2(k - \ell).$$

Then $xy = 2(k - \ell)x$. Since $(k - \ell)x \in \mathbb{Z}$, we see that xy is even.

Case 2: Assume that x is even. Then $x = 2m$ for some $m \in \mathbb{Z}$. Hence $xy = 2(my)$, and since $my \in \mathbb{Z}$, we see that xy is even. $\square$

- A. The theorem is true and the proof is correct.
- B. The theorem is false.
- C. The theorem is true, but the proof is incorrect because it assumes that x is odd.
- D. The theorem is true, but the proof is incorrect because it assumes that x is even.
- E. The theorem is true, but the proof is incorrect because it does not deal with the case where y is odd.

(b) True or false: If $a \equiv 3 \pmod{5}$, then $a^2 \equiv 4 \pmod{5}$.

(c) True or false: The converse of the open sentence “If x is even, then $x + 1$ is odd.” is the open sentence “If $x + 1$ is even, then x is odd.”

2. Let $x, y \in \mathbb{Z}$. Prove that if $x^2 - xy$ is odd, then x is odd and y is even.

3. Prove the following statement. Given real numbers $x, y, z \in \mathbb{R}$, if x and y are rational, $x \neq 0$, and z is irrational, then $\frac{y+z}{x}$ is irrational.

4. Prove the following statement (giving a *single* justification for every step you make). For any two sets S and T , then $S \cap T = S$ if and only if $S \subseteq T$.